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Information System Development for Dental Clinic Using ReactJS and Prisma ORM

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Abstract

drg. Irna Dental Clinic is a healthcare facility that still uses a semi-manual system for patient registration and medical recordkeeping, leading to various issues such as service delays, recording errors, and difficulties in managing patient data. This study aims to design and develop a web-based information system that integrates administrative processes, including patient registration, appointment scheduling, medical record management, and online consultation features. The system was developed using the Software Development Life Cycle (SDLC) Waterfall model, employing modern web technologies for the frontend, user interface design, and database management. System testing was conducted using the Black Box Testing method with 48 test cases, all of which were successfully executed as expected. This indicates that all the core features of the system function properly and consistently. The results show that the system can improve the clinic's operational efficiency, simplify access to information, and support more organized and modern patient services. By transitioning from a semi-manual system to a fully digital one, the clinic can provide faster, more accurate, and easily accessible services, significantly enhancing the overall patient experience and helping the clinic keep up with technological advancements in healthcare service delivery. Additionally, the feedback from the clinic's staff (doctors and admins) confirmed that the system meets the operational goals and provides a more efficient solution compared to the previous manual processes.

I. INTRODUCTION

In today's era of globalization, information technology is rapidly advancing, including in the healthcare sector. Health is a condition of physical and mental productivity; without it, daily activities and work can be disrupted [1]. Many sectors—from production and sales to agriculture and livestock—have utilized web and internet technologies to support their activities. By simply opening a browser, users can access the information they need to achieve their goals [2]. Healthcare facilities such as hospitals, community health centers, and clinics have begun shifting from manual, paper-based data management to more efficient electronic systems. The internet plays an important role in providing simple, fast, accurate, and precise access to information and services, facilitating users in their work [3]. Health service institutions highly depend on information and communication technology to deliver quality healthcare. Efficient information processing is a crucial factor for the success of these institutions. A good information system can support clinical workflows in various ways, contributing to better patient care [4].

drg. Irna's Dental Clinic is a healthcare service that focuses on dental and oral care. This clinic offers a variety of services, such as dental check-ups, tartar cleaning, fillings, tooth extractions, and other treatments to maintain patients' oral health. However, drg. Irna's Clinic currently still uses a conventional system in its operations. The registration process is still conducted semi-manually, where patients register via messaging applications, and then the staff manually enters the patient data into the system.

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Using the Software Development Life Cycle (SDLC) Waterfall model, this web-based system is designed to enhance effectiveness and efficiency, as evidenced by testing results showing that the system is feasible to use. To address these issues, drg. Irna's Clinic requires a web-based information system that integrates various clinic administrative processes. The system is expected to manage patient registration, appointment scheduling, medical record documentation, and structured and automated data management, as well as offer paid online consultation features. Choosing a web-based system allows easy access from various devices and locations, providing patients with flexibility to make appointments or conduct consultations online without visiting the clinic in person. Moreover, using a website as the main platform is more practical than desktop or mobile applications, as it does not require users to install additional software. A website can be accessed through browsers on various devices, whether on computers or smartphones, without being tied to a specific platform, thereby making it easier for both patients and clinic staff to access and use the system anytime and anywhere. With the implementation of a web-based system, clinic operations will become more organized, enabling staff to focus on direct patient care, reducing data errors, and improving convenience and accessibility for patients in scheduling appointments, conducting online consultations, and enabling doctors to view patient medical records more easily.

The framework used in this study is ReactJS, which offers advantages such as ease of implementation, structured components, and optimal performance in building interactive user interfaces [2], [5]. Hasnain and Ullah explain that ReactJS's reusable components facilitate easier development and maintenance, helping to overcome common programming challenges [5]. Additionally, Prisma ORM is employed to manage the database more intuitively, enabling better-organized data integration and simplifying data query and manipulation processes in a secure and efficient way [3], [6], [7]. Prisma focuses on objects within the system rather than requiring manual SQL queries, which optimizes the time needed for data processing and reduces complexity [6], [7], [8]. The integration of ReactJS and Prisma helps develop a more structured system, minimizes errors, and enhances overall system quality and reliability [9].

Therefore, ReactJS and Prisma ORM were chosen to design and develop a web-based patient service information system for drg. Irna's Dental Clinic. This system is expected to provide a solution to the clinic's issues in managing patient registration, scheduling, and medical record documentation with better structure and more accurate data management.

II. Related Works/Literature Review

Web-based information systems have become increasingly important in improving the efficiency and quality of healthcare services, particularly in clinics. Various studies have implemented such systems to address specific administrative challenges. One implementation involved the development of a dental appointment and treatment platform using the Waterfall model. The system was evaluated through black-box testing and the System Usability Scale (SUS), receiving a SUS score of 78, which confirmed that its features worked well and improved patient registration and service flow [1]. Another study developed a web-based patient registration system for a clinic setting, replacing inefficient manual processes. This implementation led to faster registration times and more organized data management [10].

Further developments targeted other aspects of clinical services. A consultation system was created for an ENT clinic to manage doctor-patient interactions more efficiently, incorporating features for data access and online consultations [11]. Similarly, a medical record system was implemented for a midwife clinic, enabling automated data management and reporting through PHP and MySQL [12]. In another case, a health center system integrated patient registration, records, and pharmacy services using VB.Net and PostgreSQL, resulting in streamlined operations and improved service quality [13].

Beyond healthcare, similar methodologies have been applied in academic and organizational environments. An information system was developed for a university program to address limitations in centralized communication, allowing structured access to academic and organizational information using PHP Laravel and JavaScript [14]. Likewise, clinics facing rising service demand reported inefficiencies due to the use of communication apps like WhatsApp. These were addressed by introducing integrated systems to reduce data errors and improve patient satisfaction [15]. Other initiatives included web-based systems for clinics in rural areas to improve information availability and reduce delays caused by manual data entry [16], [17]. Moreover, administrative systems for local organizations have also benefited from similar implementations using Laravel and MySQL, enhancing communication and record-keeping [18].

These studies consistently demonstrate the advantages of web-based systems in resolving inefficiencies, organizing clinical data, and enhancing patient service experiences. However, most existing systems focus on individual processes, such as booking, registration, consultation, or medical records, rather than integrating them into a cohesive platform. Additionally, the use of advanced web technologies like ReactJS and Prisma ORM—known for their high performance, modular design, and efficient data handling—is still limited in these applications. Therefore, a research gap remains in developing a fully integrated dental clinic information system that unifies core processes, including registration, appointment scheduling, medical records, and online

consultations, using modern web development frameworks to optimize both system performance and user experience.

III. METHODS

This study adopts the Software Development Life Cycle (SDLC) using the Waterfall model, which is structured sequentially through five main stages: communication, planning, modeling, construction, and deployment. Each phase is carried out comprehensively before proceeding to the next, ensuring systematic and complete development of the system. Figure 1 illustrates that each stage is carried out sequentially and begins only after the previous stage is completed.

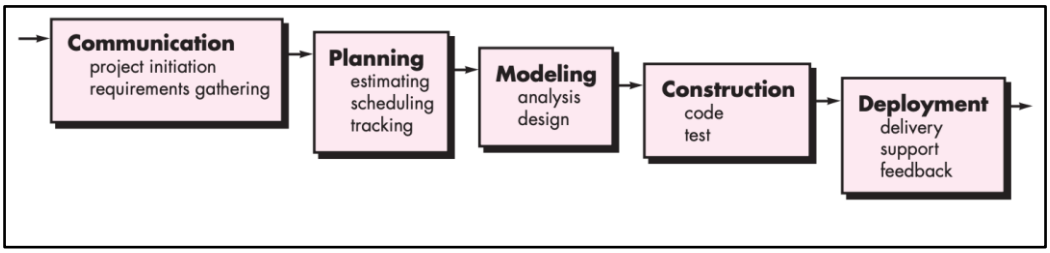


Fig. 1 Research Methods Waterfall [19]

A. Communication

In the initial stage, qualitative data were collected to identify the clinic's current administrative processes and challenges. Interviews were conducted with the clinic's main operator, drg. Irna, and observations were made on how patient registration, medical recordkeeping, and appointment scheduling were managed. The clinic's reliance on semi-manual operations highlighted several issues such as data duplication, inefficiency in patient queue handling, and absence of digital consultation features. Additionally, a literature review on similar information system implementations and the Waterfall method was carried out to support the design framework. The output of this stage is a Software Requirements Specification (SRS) that outlines the functional requirements for three main user roles: admin, doctor, and patient. The functional requirements were identified and specified based on the clinic's needs as presented in Table 1.

TABLE 1
ADMIN & DOCTOR FUNCTIONAL REQUIREMENTS

No	Admin Page	Feature	Description
1	Authentication	Login / Logout	Admin and doctors log in/out using registered accounts.
2	Dashboard	View Admin Dashboard	Displays total patients, appointment count, and latest treatment fees.
3	Appointment Data	View Booked Appointments	Lists all appointments booked by patients.
		Accept/Reject Appointment	Admin can approve or reject appointment requests.
		Update Payment Method	Admin updates the payment method details for appointments.
4	Patient Data	View All Patients	Displays a list of all registered patients.
		Add/Edit/Delete Patient	Admin can manage patient records (add, edit, delete).
5	Medical Records	View Medical Records	Displays all stored patient medical records.
		View Details	Shows detailed information of selected records.
		Add/Edit/Delete Records	Admin or doctor can manage patient records.
		Generate PDF	Converts medical records to PDF for printing or archiving.
6	Chat Consultation (Doctor)	View Chat Sessions	Shows consultation chat history between doctor and patient.
		Start/End Session	Doctor begins and ends chat session as per schedule.
		Conduct Chat	Enables direct communication in consultation session.
7	Doctor Schedule	View Schedule	Displays doctor practice schedules.
		Add/Edit/Delete Schedule	Admin can manage doctor availability.
8	Treatment Types	View Treatment Types	Lists available medical treatment types.
		Add/Edit/Delete Treatment Type	Admin can manage treatment list and update fees.

B. Planning

In the planning stage, a comprehensive project schedule was developed to guide the system development activities. This process began with breaking down the project into detailed tasks and estimating the time and

resources required for each. A Gantt chart was used as the primary planning tool to visualize the project timeline, task dependencies, and milestones. The planning also involved risk identification and mitigation strategies to ensure the project stayed on track. Regular coordination with the clinic's representative was conducted to align priorities and confirm the feasibility of the planned timeline. This stage was essential to minimize potential delays, optimize the allocation of human and technical resources, and provide a structured workflow for the development team.

C. Modeling

The system architecture and interface were designed using Object-Oriented Design (OOD) principles. Modeling tools included UML diagrams such as use case diagrams (to map user-system interactions), activity diagrams (to illustrate workflow sequences), class diagrams (to define system objects and their relationships), and sequence diagrams (to represent communication flows). Wireframes were also designed to plan the layout and navigation of the user interface. These designs ensured the system would be user-friendly and aligned with the operational needs of clinic staff and patients.

1. Use Case Diagram
   Shows how the system interacts with users or other systems, which helps to clearly and easily understand the system's needs or requirements [20].
2. Activity Diagram
   The activity diagram models the flow of activities within a system, showing how processes start, decisions made, and how they end. It provides a high-level overview of the process, including possible parallel activities, without detailing internal system behavior. Usually based on one or more use cases, it helps illustrate the sequence of business processes and resembles a flowchart or DFD [21].
3. Sequence Diagram
   The primary purpose is to document and illustrate the classes to be developed in programming. The design class diagram represents the necessary object-oriented classes, their navigation relationships, attribute names and properties, as well as method names and their properties [22].
4. Class Diagram
   The primary purpose is to document and illustrate the classes to be developed in programming. The design class diagram represents the necessary object-oriented classes, their navigation relationships, attribute names and properties, as well as method names and their properties [23].

D. Construction

The system was built using modern web technologies. The frontend was developed using ReactJS with Tailwind CSS for styling and Redux for state management. Backend services were built using ExpressJS, connected to a PostgreSQL database, and managed with Prisma ORM for efficient data handling. APIs were developed to handle communication between frontend and backend components. The system features included digital patient records, doctor scheduling, admin dashboards, and online chat-based consultation. Testing was carried out through Black Box Testing and User Acceptance Testing (UAT) involving clinic staff and patients.

1. Black Box Testing
   Black box testing is a software testing technique in which the tester evaluates the system solely based on its inputs and expected outputs, without any knowledge of its internal structure or code. Test cases are created according to the software's functional requirements and specifications, ensuring that the system behaves as intended and meets user expectations [24].
2. User Acceptance Testing (UAT)
   User Acceptance Testing (UAT) is a software testing process conducted by end users before the new information system is implemented in an organization. Its main purpose is to ensure that the system meets its original development goals and aligns with defined business needs [25].

E. Deployment

Upon successful testing, the system was deployed and handed over to the clinic administrator. A training session was conducted to guide the clinic staff on using the system. Feedback from users was collected during the post-deployment stage to identify potential improvements. The system was hosted on a web server to ensure accessibility via desktop and mobile browsers without requiring installation.

IV. RESULTS

The development process resulted in a web-based information system tailored to support the administrative and medical operations of Klinik drg. Irna. The system consists of three main user roles: admin, doctor, and patient, each with distinct access rights and functionalities. Key features include online appointment scheduling, digital medical records, doctor scheduling management, and chat-based consultations.

A. Use Case Diagram

Figure 2 illustrates the use case diagram for the information system of Klinik drg. Irna. The diagram depicts the primary interactions between users and the system, showcasing the roles of admin, doctor, and patient. The system is designed to streamline the clinic's operations by integrating several key functions.

The admin role is responsible for managing appointments, patient data, doctor schedules, and medical records. Additionally, the admin can generate reports and oversee consultations. The doctor role includes conducting consultations, managing their schedule, and providing consultations. Doctors are also able to print reports for patients. Meanwhile, patients can register for consultations, choose an online consultation schedule, view appointment histories, and make payments for services. This diagram reflects how each user interacts with the system to improve efficiency and organization in the clinic's operations, from registration to consultation and medical record management.

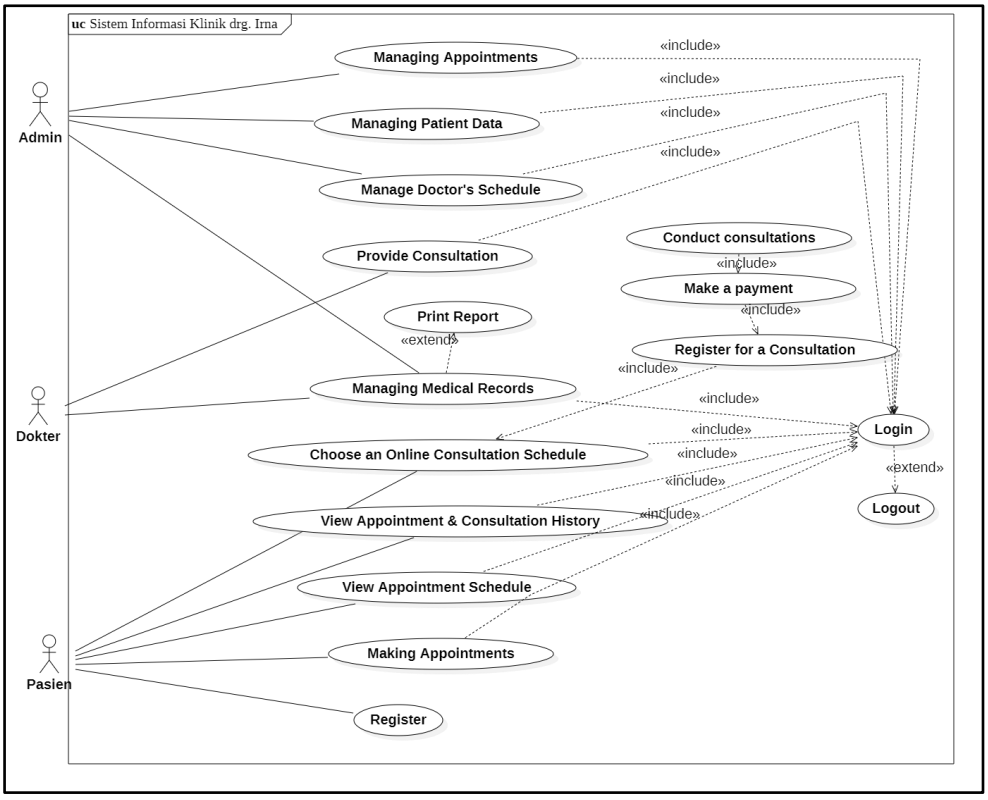


Fig. 2 Use Case Diagram

B. Activity Diagram

Figure 3 illustrates the activity diagram for the Booking Appointment process in the Klinik drg. Irna's information system. This diagram details the sequence of steps that a patient follows to schedule an appointment with the clinic, as well as the corresponding interactions with the system. The process begins when the patient logs in to the system, which then displays the landing page. After logging in, the patient selects the appointment menu. The system presents the available consultation schedule for the next month, allowing the patient to either directly choose an available appointment slot or use a date filter to narrow down the options. Following the appointment selection, the patient enters a short complaint, which is a brief description of the issue they are seeking consultation for. This helps the doctor prepare for the consultation in advance. After this step, the patient confirms the appointment selection. If the patient proceeds with the confirmation, the system waits for admin approval for 24 hours. Once the appointment is confirmed, the patient receives a notification indicating that the appointment has been successfully confirmed. In the event the patient cancels the appointment or if the appointment cannot be confirmed, a cancellation notification is sent to the patient.

This diagram provides a clear view of how the system processes appointment bookings, from user input to final confirmation, and how both successful and unsuccessful attempts are handled.

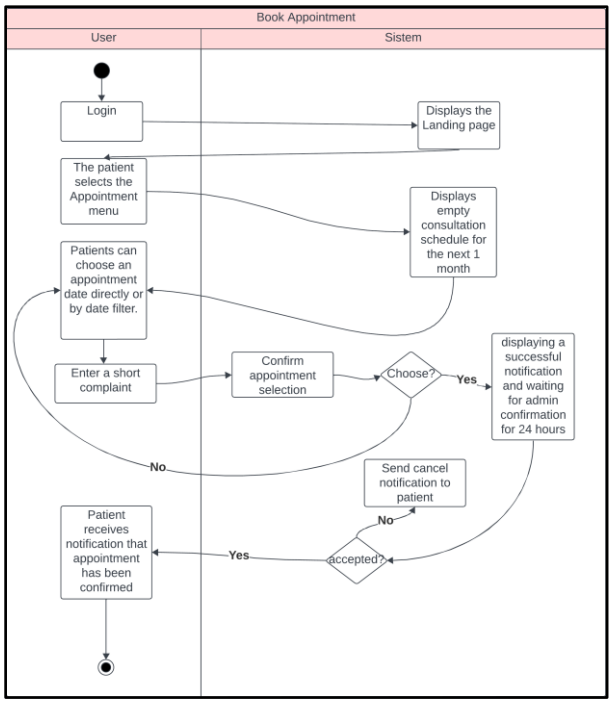


Fig. 3 Activity Diagram Book Appointment

C. Sequence Diagram

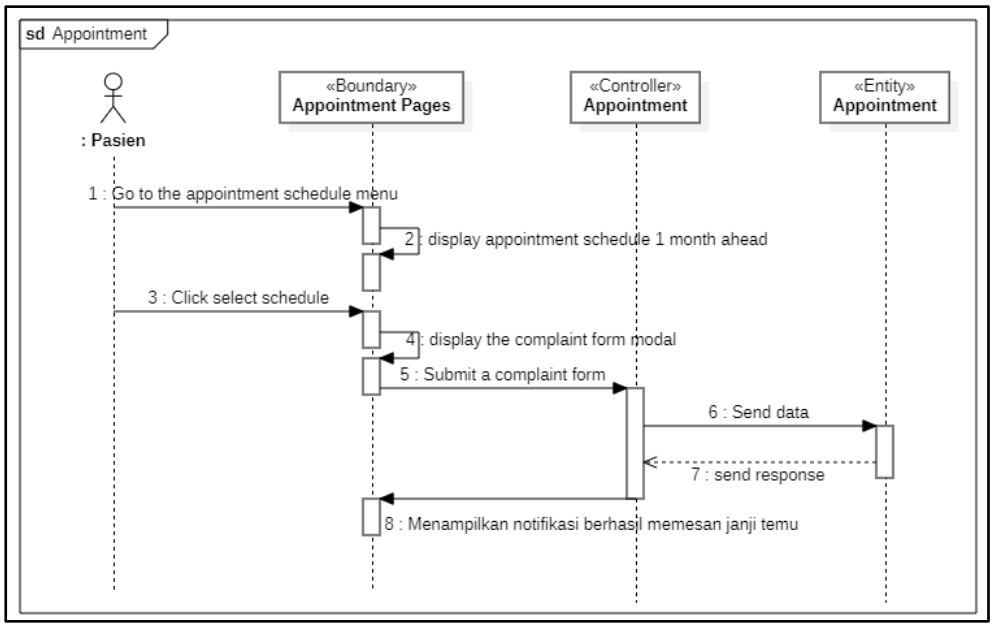


Fig. 4 Sequence Diagram Appointment

Figure 4 illustrates the sequence diagram for the Appointment Booking process in the Klinik drg. Irna's information system. This diagram outlines the interaction between different components of the system when a patient books an appointment. The diagram involves three main components: the Appointment Pages (Boundary), the Appointment (Controller), and the Appointment Entity (Entity). The process begins when the Appointment Pages (Boundary) component receives a request to initiate the booking process. It then communicates with the Appointment controller, which is responsible for managing the flow of data and operations. The controller verifies the user's input and proceeds with the logic required for the booking process, such as checking available time slots and confirming the patient's details. After the controller processes the input, it interacts with the Appointment Entity, which handles the data storage related to the appointment, such as saving patient details, appointment time, and other relevant information. Once the appointment is successfully stored, the system returns

the appropriate response to the Appointment Pages component, confirming that the booking was successful or informing the patient of any issues.

This diagram helps to visualize the step-by-step communication between the system's components, ensuring that the appointment booking process is performed correctly.

D. Class Diagram

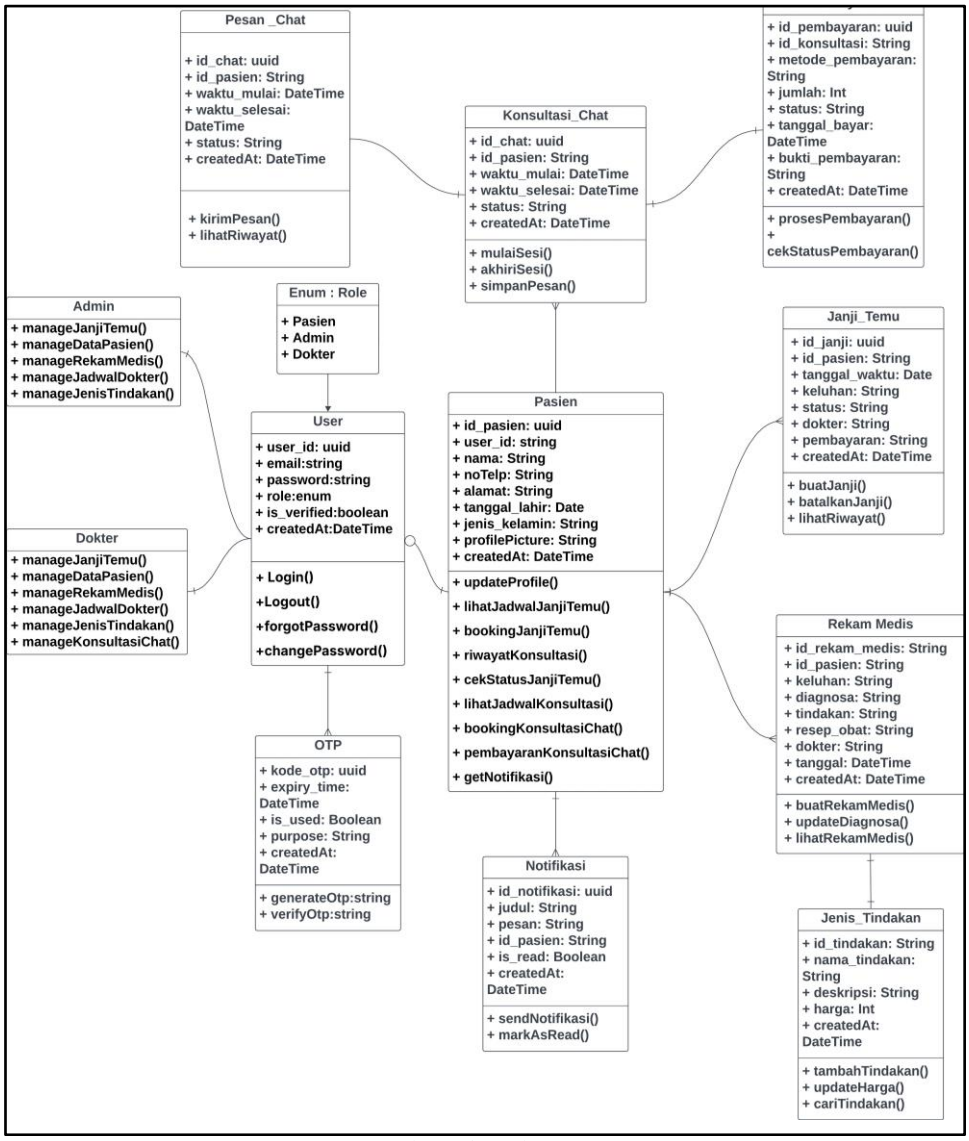


Fig. 5 Class Diagram

Figure 5 illustrates the class diagram of the information system for Klinik drg. Irna, showing key classes such as Pesan_Chat, Consultasi_Chat, User, Pasien, Dokter, Janji_Temu, Rekam_Medis, Jenis_Tindakan, OTP, and Notifikasi. Each class represents a specific function within the system, such as managing communication between patients and doctors (Pesan_Chat and Konsultasi_Chat), handling user authentication (User), storing patient and appointment data (Pasien, Janji_Temu), and managing medical records (Rekam_Medis). The Jenis_Tindakan class defines treatment details, while OTP ensures secure user verification and Notifikasi manages notifications. This diagram provides a clear structure of how the system components interact to streamline clinic operations.

E. Black Box Testing

A total of 48 test cases were executed during Black Box Testing. All of them passed successfully, which confirmed that the system's core functionalities (such as appointment scheduling, patient registration, medical record management, and consultation) functioned properly and as expected.

Table 2 presents the Appointment Testing for Patients, focusing on the patient's ability to view available doctor schedules, select a schedule, and submit an appointment request. All test cases passed successfully, confirming that patients can easily access the appointment system, choose their preferred time slot, and submit the necessary

data. The system ensures appointments are saved with a "pending" status, providing a clear and organized process for both patients and the clinic.

TABLE 2
APPOINTMENT TESTING FOR PATIENTS

No	Test Case	Expected Result	Test Result	Status
1	P_JanjiTemu_ViewSchedule_01	Patient views the complete list of doctor schedules	The system displays a table with the doctor's schedule (date, day, time)	Success
2	P_JanjiTemu_SelectSchedule_01	Patient selects a doctor's schedule to make an appointment	The system displays the appointment form based on the selected schedule	Success
3	P_JanjiTemu_Submit_01	Patient fills out and submits an appointment request	The system saves the appointment data with a "pending" status and shows a notification	Success

F. User Acceptance Test (UAT)

The User Acceptance Testing (UAT) for the Klinik drg. Irna information system was conducted by the doctor role. The UAT covered several modules, including Authentication, Appointment Management, Patient Data Management, Medical Record Management, Consultation Chat Management, Doctor Schedule Management, and Treatment Type Management. The tests were performed on 31 May 2025, with each module being thoroughly evaluated. Based on the results shown in Table 3, all modules successfully passed the testing phase, confirming that the system functions as expected and meets user requirements.

TABLE 3
UAT RESULT

No	Module	Status/Comment	Test Date
1	Authentication	Success	May 31, 2025
2	Appointment Management	Success	May 31, 2025
3	Patient Data Management	Success	May 31, 2025
4	Medical Records Management	Success	May 31, 2025
5	Consultation Chat Management	Success	May 31, 2025
6	Doctor Schedule Management	Success	May 31, 2025
7	Treatment Type Management	Success	May 31, 2025
8	Latest Consultation Chat List	Success	May 31, 2025

G. System Display

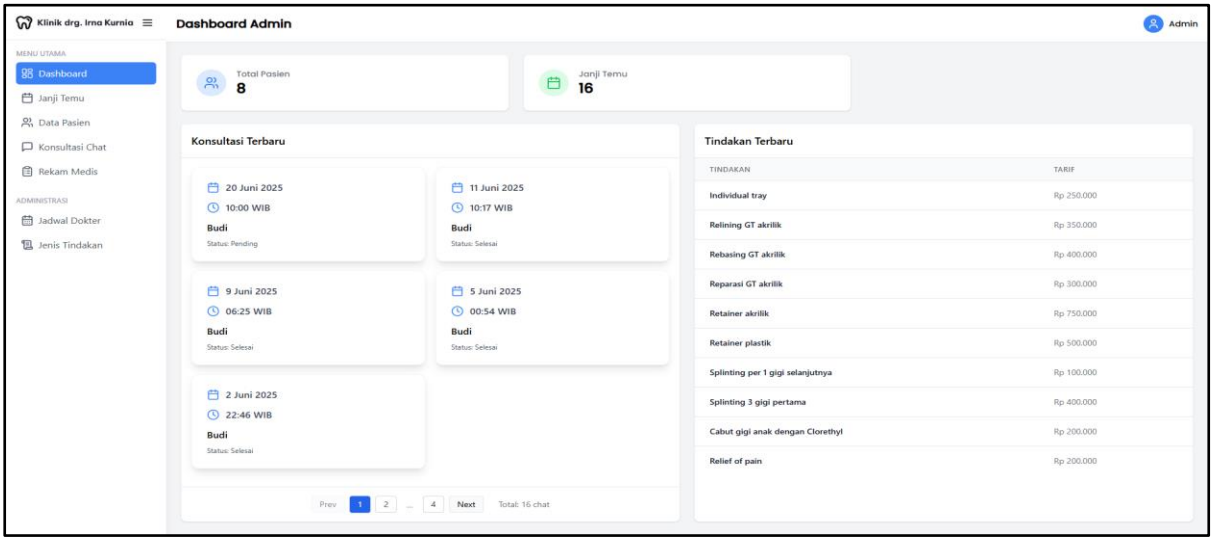


Fig. 6 Dashboard Admin

Fig. 6 displays the admin dashboard of the system, providing an overview of key clinic metrics. It includes the total number of patients, total appointments, and the latest consultations, which are shown with the patient's name, consultation date, and status. Additionally, the dashboard lists recent treatments and their respective fees, offering an efficient way for the admin to track and manage clinic activities.

V. DISCUSSION

The development of the Information System for Dental Clinic has successfully enhanced the clinic's operational efficiency and improved the patient experience. The system has streamlined processes like patient registration, appointment scheduling, medical record management, and online consultations. However, there are areas that require further attention.

While the system implements encryption to protect patient data, further testing on security and privacy has not yet been conducted in-depth. To improve patient data protection, future development should focus on end-to-end encryption and stronger communication protocols, especially during online consultations.

Currently, the system uses role-based access restrictions for limiting user access. While this works for a small number of users, additional testing is needed to evaluate the system's performance with larger user bases. Performance metrics, such as system response time and database efficiency, should be examined under heavier loads to ensure the system remains efficient as usage grows.

Although the system was tested using Black Box Testing and User Acceptance Testing (UAT), further insights into performance metrics such as response times and user feedback on ease of use would provide a more comprehensive understanding of the system's real-world performance. Overall, feedback from clinic staff and patients has been positive, with praise for the system's user-friendly interface and easy access to important features.

To further enhance the user experience, integrating real-time notifications would allow patients and clinic staff to receive instant updates on schedule changes, appointment confirmations, and test results. This would help improve communication and decision-making in the clinic.

Although ReactJS and Prisma ORM have proven effective, exploring other technologies, such as Angular for the frontend or traditional database management systems for the backend, could offer new insights into the strengths and weaknesses of the chosen technologies. These comparisons would help guide future decisions on system architecture.

Finally, improving data security is essential for ensuring patient trust. Future development should prioritize additional encryption methods and more robust network security, particularly as the system scales. Incorporating AI-driven features, such as predictive scheduling based on patient behavior, could also optimize resource allocation and improve the overall patient experience.

VI. CONCLUSIONS

The Klinik drg. Irna's Information System effectively integrates the clinic's administrative processes into a streamlined, web-based platform. This system addresses key operational challenges, including slow registration, data duplication, and inefficient appointment scheduling. By transitioning from a semi-manual to a fully digital system, the clinic has significantly enhanced both operational efficiency and patient service delivery.

This research contributes to the field of healthcare information systems by demonstrating how modern web technologies, such as ReactJS and Prisma ORM, can be utilized to improve clinic management. The successful implementation of the system, along with the positive feedback from clinic staff and patients, confirms that it meets the clinic's needs and provides a strong foundation for future upgrades.

To ensure the system continues to evolve and meet the growing demands of healthcare service delivery, future research should focus on expanding the system's capabilities. This could include advanced features such as real-time notifications, AI-powered scheduling, and integration with other healthcare platforms, which would further enhance efficiency and service quality for both patients and clinic staff.

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